AlphaQuant

Impulse Flow Alpha AI Agent

A2A-Compliant Quantitative Trading Strategy Analysis Service

Submission Details

Hackathon	Nasiko AI Agent Hackathon 2026
Track	Beat The Market — BTC-USDT
Organiser	SNTC & COPS, IIT (BHU) Varanasi
Team	Codex
Institute	Indian Institute of Technology (BHU), Varanasi
Year	2026

Abstract

AlphaQuant is a production-grade, modular AI agent that encapsulates the *Impulse Flow Alpha* quantitative trading strategy inside a fully compliant Nasiko A2A JSON-RPC 2.0 service. Built on a clean three-layer architecture — a core analytical toolset, a keyword-routing agent, and an A2A-compliant executor — the system exposes five structured tools: strategy explanation, performance analysis, risk evaluation, live signal generation, and full backtest execution. The underlying strategy was backtested on 1,462 days of BTC-USDT daily OHLC data from 2020 to 2024, producing a net return of **4307.12%**, an annualised CAGR of **157.32%**, a Sharpe ratio of **1.97**, and a maximum drawdown of **30.84%**. Robustness was validated through four independent stress tests. All five tools were verified live against the Nasiko gateway, producing fully spec-compliant A2A responses with unique context IDs, artifact IDs, UTC timestamps, and proper error envelopes. The agent deploys with a single Docker command and is extensible by design.

"Don't read the strategy. Ask it."

Contents

1. Introduction

AlphaQuant is a purpose-built artificial intelligence agent submitted to the Nasiko AI Agent Hackathon, organised by the Science and Technology Council (SNTC) and COPS at IIT (BHU) Varanasi. The hackathon challenge — Beat The Market — required teams to develop algorithmic trading strategies on BTC-USDT daily price data and demonstrate capability design, A2A gateway compliance, code reliability, and output predictability.

AlphaQuant satisfies all four requirements through a rigorously engineered system. It is not a chatbot, not a prompt wrapper, and not a single-call script. It is a three-layer modular Python service that encapsulates the entire intellectual content of the *Impulse Flow Alpha* trading strategy — its signal logic, backtest engine, stress test findings, and risk analysis — inside a conversational AI interface accessible through the Nasiko A2A JSON-RPC 2.0 protocol.

Core Properties

- **Five structured tools** covering the complete strategy analysis lifecycle
- **Fully deterministic** — identical input always produces identical output
- **Zero hardcoded responses** in Tools 4 and 5 — all results computed live from raw data
- **Full A2A compliance** — tested live, all gateway fields verified
- **One-command deployment** via Docker Compose
- **Dual-format data ingestion** — accepts JSON arrays and CSV text blocks

2. Problem Statement

Quantitative trading strategies are typically encoded as Jupyter notebooks or Python scripts containing numerical thresholds, signal logic, and backtest loops. This representation is powerful for development but fundamentally inaccessible for communication, querying, and integration.

2.1. The Accessibility Gap

A practitioner, team member, or automated system wishing to ask any of the following questions must manually run code and interpret raw numerical outputs:

- Should I go long on this asset right now?
- Is this strategy robust to parameter changes?
- What do these backtest metrics actually mean in context?
- How did the strategy perform across the full four-year dataset?

There is no standardised interface between a strategy's internal logic and external consumers. AlphaQuant bridges this gap entirely.

2.2. The Solution

AlphaQuant wraps the complete Impulse Flow Alpha strategy inside a conversational AI service. A plain English query produces a precise, structured, professionally formatted analytical response in milliseconds, through a standardised endpoint, with full A2A protocol compliance. The complexity of the Python code is abstracted away while preserving complete mathematical rigour.

3. The Impulse Flow Alpha Strategy

3.1. Conceptual Intuition

Impulse Flow Alpha is a pattern-recognition-based long/short strategy built on the empirical observation that price frequently moves in a predictable three-phase structure across trending assets:

1. **IMPULSE** — A strong, sharp directional price move on a single bar that signals directional intent and identifies a potential setup anchor.

2. **CONSOLIDATION** — A shallow pullback or sideways pause in the bars following the impulse. The structure established by the impulse must remain intact — excessive retracement invalidates the setup.

3. **BREAKOUT** — A continuation move in the direction of the original impulse, confirming that the prior structure was a valid pause rather than a reversal.

The strategy enters a trade *only when all three phases are precisely confirmed*, significantly reducing false signals and improving the quality and selectivity of each entry.

3.2. Signal Parameters

Table 1 lists all eleven strategy parameters and their values for both long and short setups. These constants are defined at the top of `agent_toolset.py` and are referenced by all five tools, ensuring mathematical consistency throughout the system.

Table 1: Impulse Flow Alpha — Signal Parameters

tablehead		
Impulse bar return (min)	$+0.13\%$	-7.0%
Impulse bar return (max)	$+0.27\%$	-2.0%
Maximum retrace	3.0%	1.0%
Breakout range (min)	$+1.9\%$	-6.30%
Breakout range (max)	$+5.0\%$	-5.80%
Lookback window	40 bars	
Transaction cost	0.15% per trade	

3.3. Asymmetric Design Rationale

The long and short setups are intentionally asymmetric. The long impulse window (0.13%–0.27%) is deliberately narrow, capturing small but meaningful one-day moves that precede sustained uptrends in Bitcoin. The short setup uses a much wider and more negative impulse range (-2.0% to -7.0%), reflecting the empirical observation that downside moves in cryptocurrency tend to be sharper and more abrupt than upside moves.

This asymmetry gives the strategy genuinely different sensitivities to bullish and bearish market regimes — a property that is analytically important and that AlphaQuant's `explain_strategy` tool communicates explicitly.

3.4. Backtest Performance

The strategy was backtested on 1,462 days of BTC-USDT daily OHLC data from January 2020 to December 2024, with a starting capital of $1,000,000 and a flat transaction cost of 0.15% per trade. Results are presented in Table 2.

Table 2: Backtest Performance — BTC-USDT (Jan 2020 – Dec 2024)

tablehead	
Net Return	**4307.12%**
Gross Profit	4759.13%
Gross Loss	−452.01%
Annualised CAGR	**157.32%**
Sharpe Ratio	**1.9659**
Sortino Ratio	**3.0001**
Calmar Ratio	**5.1007**
Maximum Drawdown	30.84%
Starting Capital	$1,000,000
Final Capital	∼$44,071,246
Total Days	1,462
Round-trip Trades	14

3.5. Stress Testing

Four independent stress tests were conducted to validate strategy robustness beyond the primary backtest result. These tests are summarised in Table 3 and detailed below.

3.5.1. Test 1 — Trade Connectivity (20% Random Dropout)

In each of 1,000 simulations, 20% of all trades were randomly removed and the PnL of the remaining 80% was computed. The average simulated annualised return was approximately 114%, compared to the original 157.32%. This confirms that alpha is broadly distributed across the trade set and not concentrated in a small number of fortunate entries.

3.5.2. Test 2 — Monte Carlo Sequence Shuffle

The order of all 14 round-trip trades was shuffled randomly across 2,000 simulations and the maximum drawdown was computed for each sequence. The actual maximum drawdown (30.84%) was lower than the average shuffled drawdown (41.74%) and the 95$^{\text{th}}$ percentile VaR drawdown (55.70%). This provides strong evidence that the strategy's trade timing is genuinely skilful rather than coincidentally favourable.

3.5.3. Test 3 — Gaussian Diffusion Baseline

The mean daily return μ and daily volatility σ were estimated from the strategy's equity curve and used to parameterise a lognormal diffusion process. Thousands of synthetic price paths

were simulated and their total returns were compared to the actual result. The real equity curve exceeded the 95th percentile of the simulated distribution, confirming that the returns are not attributable to random price drift alone.

3.5.4. Test 4 — Parameter Perturbation

Each of the eleven strategy parameters was independently perturbed by $\pm 5\%$ and the resulting returns and drawdowns were recorded (Table 4). The majority of parameters showed zero sensitivity. The most sensitive parameter was `breakout_max_short`: a $+5\%$ change reduced net returns by approximately 562 percentage points. This parameter warrants monitoring in live deployment.

Table 3: Stress Test Summary

tablehead		
Trade Dropout (20%)	Avg return $\sim$114%	Alpha distributed
MC Shuffle (2000 iter)	Actual DD $<$ avg DD	Timing genuine
Gaussian Diffusion	Exceeds 95th pct band	Not luck-driven
Param Perturbation	Most params zero sens.	Robust

Table 4: Parameter Perturbation Results ($+5\%$ change)

tablehead		
`impulse_min_long`	4307.12	30.84
`impulse_max_long`	4764.29	32.02
`breakout_min_long`	4307.12	30.84
`breakout_max_long`	4282.36	31.23
`retrace_max_long`	4307.12	30.84
`impulse_min_short`	4307.12	30.84
`impulse_max_short`	4307.12	30.84
`breakout_min_short`	4129.53	35.36
`breakout_max_short`	3744.87	30.84
`retrace_max_short`	4307.12	30.84
`window_threshold`	4307.12	30.84

4. System Architecture

4.1. Three-Layer Design

AlphaQuant follows a clean three-layer architecture with strict separation of concerns. Each layer has a single responsibility and is independently testable and replaceable.

```
  User Query / Nasiko Gateway
          |
          v  [POST /  JSON-RPC 2.0]
+--------------------------------+
|  Layer 3     A2A Executor      |
|  openai_agent_executor.py      |
|  - Parses JSON-RPC 2.0 input   |
|  - Delegates to agent          |
|  - Wraps output in A2A format  |
+--------------------------------+
          |
          v
+--------------------------------+
|  Layer 2     Agent Router      |
|  openai_agent.py               |
|  - Keyword intent detection    |
|  - GPT-4o-mini LLM fallback    |
|  - JSON + CSV data extraction  |
+--------------------------------+
          |
          v
+--------------------------------+
|  Layer 1     Core Toolset      |
|  agent_toolset.py              |
|  - 5 async tools               |
|  - Zero external API calls     |
|  - NumPy / Pandas computations |
+--------------------------------+
          |
          v
  FastAPI App (__main__.py)
  POST /  |  GET /.well-known/agent.json  |  GET /health
```

Listing 1: AlphaQuant System Architecture

4.2. Layer 1 — The Toolset (`agent_toolset.py`)

The toolset is the core analytical engine and the largest file in the project (14 KB). It contains all five tools as `async` Python methods inside the `QuantTradingToolset` class. This layer has zero dependency on any external API and can operate entirely offline.

Key design properties:

- All eleven strategy parameters are module-level constants, referenced by every tool to ensure mathematical consistency
- The signal engine in `generate_signal` is a direct, faithful port of the `daily_signal` function from the original backtester notebook
- The backtest loop in `run_backtest` is a direct port of the `backtest` function, producing numerically identical results
- All metric computations (Sharpe, Sortino, Calmar, drawdown, win rate, profit factor) are implemented from scratch using NumPy — no third-party financial libraries
- Comprehensive input validation and data cleaning in every tool that accepts

external data

4.3. Layer 2 — The Agent (`openai_agent.py`)

The agent layer handles intent detection and LLM fallback. The intent router (`_detect_intent`) is a priority-ordered keyword matcher evaluated in a deliberate sequence:

1. `run_backtest` — checked first to prevent conflicts when a message contains both data and metric-related keywords
2. `explain_strategy`
3. `analyze_performance`
4. `evaluate_risk`
5. `generate_signal`
6. LLM fallback (GPT-4o-mini, temperature 0.3, max 1024 tokens)

The `_extract_ohlc` utility can parse OHLC data from two formats embedded anywhere in the user's message:

- **JSON array** — scans for outermost `[...]` delimiters and parses with `json.loads`
- **CSV block** — identifies comma-containing lines, extracts them, parses with `csv.DictReader`, validates required columns case-insensitively, and normalises keys to title case

When 30 or more bars are supplied with a signal keyword, the agent automatically escalates the request to `run_backtest`, recognising that a large dataset is more valuable as a full simulation.

4.4. Layer 3 — The Executor (`openai_agent_executor.py`)

The executor is the A2A compliance boundary. It performs three functions:

1. **Input parsing** — extracts the user message from the nested `params.message.parts` structure with defensive error handling
2. **Delegation** — calls `OpenAIAgent.run()` and awaits the response string
3. **Response wrapping** — constructs a fully spec-compliant A2A result envelope with all required fields

Every failure is caught and returned as a clean JSON-RPC 2.0 error envelope with a numeric error code, message string, and UTC timestamp — never a raw Python traceback.

4.5. FastAPI Application (`__main__.py`)

The three layers are served by a FastAPI application exposing three HTTP endpoints:

5. Project File Structure

```
AlphaQuant/
|
+-- src/
|   +-- __init__.py                 # Package initialiser
|   +-- __main__.py                 # FastAPI app (3 KB)
|   +-- agent_toolset.py            # All 5 tools (14 KB)
```

Table 5: HTTP Endpoints

tablehead		
POST	/	Primary A2A endpoint
GET	/.well-known/agent.json	AgentCard discovery
GET	/health	Liveness probe

```
|   +-- openai_agent.py              # Intent router + LLM (6 KB)
|   +-- openai_agent_executor.py     # A2A compliance layer (5 KB)
|
+-- AgentCard.json                   # Nasiko registry metadata
+-- Dockerfile                       # Python 3.11-slim container
+-- docker-compose.yml               # One-command deployment
+-- pyproject.toml                   # Dependency specification
+-- README.md                        # Setup and usage guide
+-- CUSTOMIZE.md                     # Extension guide
```

Listing 2: Complete Project File Structure

6. The Five Tools

6.1. Tool 1: `explain_strategy`

Trigger keywords: *explain, how does, what is impulse, describe strategy*

Input: Optional `strategy_text` string (additional context)

Output: Structured multi-section report covering:

- Three-phase ICB conceptual intuition
- Exact numerical thresholds for long and short setups, read live from module-level constants
- Execution assumptions: lookback window, transaction cost, position types
- Optional additional context from the user message

Key property: Values are read directly from strategy constants, not from hardcoded strings, ensuring the explanation is always mathematically consistent with the actual signal engine.

6.2. Tool 2: `analyze_performance`

Trigger keywords: *performance, metrics, sharpe, sortino, calmar, annualised, drawdown*

Input: Seven numerical parameters with official BTC-USDT backtest results as defaults (callable with zero arguments)

Qualitative rating thresholds:

- Sharpe: ≥ 2.0 = Excellent, ≥ 1.5 = Good, ≥ 1.0 = Acceptable, < 1.0 = Poor
- Sortino: ≥ 3.0 = Excellent, ≥ 2.0 = Good

- Calmar: ≥ 3.0 = Excellent, ≥ 2.0 = Good
- Drawdown: rated as Low / Moderate / High / Very High risk

Output: Return metrics, risk-adjusted metrics with ratings, drawdown assessment, and contextual interpretation commentary.

6.3. Tool 3: `evaluate_risk`

Trigger keywords: *risk, stress, robust, overfitting, monte carlo, perturbation*

Input: None

Output: Structured report covering:

- All four stress test results with specific numerical findings
- Five forward-looking risks: small sample size, in-sample only evaluation, BTC-specific calibration, absence of live slippage modelling, regime dependency
- Conclusion with recommended next steps

6.4. Tool 4: `generate_signal`

Trigger keywords: *signal, generate signal, ohlc, predict, trade now, go long, go short*

Input: JSON array of OHLC dicts (minimum 5 bars)

Processing pipeline:

1. Validate required columns (Date, Open, High, Low, Close)
2. Cast to numeric, replace zero/negative with NaN
3. Forward-fill missing values, drop remaining NaN rows
4. Run signal engine on final bar (identical to backtester logic)

Output: Signal label (LONG, SHORT, FLAT) with trigger detail string identifying the specific bar index, return %, retrace %, and breakout % that satisfied the entry conditions.

Auto-escalation: Automatically routes to `run_backtest` when 30 or more bars are supplied.

6.5. Tool 5: `run_backtest`

Trigger keywords: *backtest, run backtest, full backtest, simulate, test this data*

Input: OHLC data as JSON array **or** CSV text block (minimum 10 bars, any asset)

Computed metrics:

- Net return and final capital
- Annualised CAGR using exact compounding formula
- Sharpe ratio (annualised, rf $= 0$, ddof $= 1$, factor $\sqrt{365}$)
- Sortino ratio (downside returns only, ddof $= 1$)
- Calmar ratio (CAGR / Max Drawdown)
- Maximum drawdown via peak-tracking loop

- Win rate, gross profit, gross loss, profit factor
- Average winning trade %, average losing trade %

Output: Full report with input summary, all computed metrics with ratings, last 10 trade events (OPEN/CLOSE) with date, direction, price, PnL%, and running capital, and a dynamic interpretation section whose commentary adapts based on actual computed values.

Table 6: All Five Tools — Summary

tablehead

1	`explain_strategy`	Yes	ICB pattern explanation
2	`analyze_performance`	Yes	Metrics with ratings
3	`evaluate_risk`	Yes	Stress tests + risks
4	`generate_signal`	Yes	LONG / SHORT / FLAT
5	`run_backtest`	Yes	Full metrics + trade log

7. A2A Protocol Compliance

7.1. Request Schema

Every request to AlphaQuant follows the Nasiko A2A JSON-RPC 2.0 specification:

```
{
  "jsonrpc": "2.0",
  "id":      "req-001",
  "method":  "tasks/send",
  "params": {
    "id": "task-001",
    "message": {
      "role": "user",
      "parts": [
        {
          "kind": "text",
          "text": "Explain the Impulse Flow Alpha strategy."
        }
      ]
    }
  }
}
```

Listing 3: A2A Request Format

7.2. Response Schema

```
{
  "jsonrpc": "2.0",
  "id":      "req-001",
  "result": {
    "id":        "task-001",
    "contextId": "0d3a5439-5ffc-463f-81ea-a36c502cb465",
    "kind":      "task",
```

```
    "status": {
      "state":     "completed",
      "timestamp": "2026-03-29T22:24:24.522514+00:00"
    },
    "artifacts": [
      {
        "artifactId": "db678365-5fb7-4c7d-87d2-4faa77b32281",
        "name":       "alpha_quant_response",
        "parts": [
          {
            "kind": "text",
            "text": "=== Impulse Flow Alpha - Strategy Explanation
               ===\n..."
          }
        ]
      }
    ]
  }
}
```

Listing 4: A2A Response Format

7.3. Error Response Schema

```
{
  "jsonrpc": "2.0",
  "id":      "req-001",
  "error": {
    "code":    -32000,
    "message": "Execution error: <description>",
    "data": {
      "timestamp": "2026-03-29T22:24:24+00:00"
    }
  }
}
```

Listing 5: A2A Error Envelope

7.4. Compliance Verification

All five tools were verified live against the Nasiko gateway. Every response contained all required fields:

- `jsonrpc: "2.0"`
- Matching request `id`
- Unique `contextId` UUID
- Unique `artifactId` UUID
- `status.state: "completed"`
- UTC ISO-8601 `status.timestamp`
- `artifacts` array with `kind: "text"` parts
- Named artifact `alpha_quant_response`

8. Deployment

8.1. Prerequisites

- Docker Desktop 4.0+ with Compose V2
- OpenAI API key
- Port 10000 available

8.2. One-Command Deployment

```
# Windows PowerShell
$env:OPENAI_API_KEY="sk-your-key"; docker compose up --build -d

# Linux / Mac
OPENAI_API_KEY=sk-your-key docker compose up --build -d
```

Listing 6: Docker Deployment

8.3. Dockerfile

The Dockerfile is based on `python:3.11-slim` and installs all dependencies directly by name — avoiding the `pip install -e .` pattern which requires a build backend and can fail in restricted environments:

```
FROM python:3.11-slim
WORKDIR /app
RUN pip install --no-cache-dir --upgrade pip setuptools wheel
RUN pip install --no-cache-dir \
    fastapi "uvicorn[standard]" openai pandas numpy python-dotenv
COPY src/ ./src/
COPY AgentCard.json .
EXPOSE 10000
CMD ["python", "-m", "src"]
```

Listing 7: Dockerfile

8.4. Verifying the Deployment

```
# PowerShell
Invoke-RestMethod -Uri "http://localhost:10000/health"
# Expected: status=ok, agent=AlphaQuant
```

Listing 8: Health Check

9. Live Demo Results

All five tools were executed live and produced valid, fully compliant A2A responses. Table 7 summarises the demo results.

Note on Zero-Trade Results

Tools 4 and 5 correctly returned zero trades and FLAT signals for the small sample datasets used in the demo. This is the mathematically correct result — the strategy is highly selective and requires precise satisfaction of all three ICB conditions. Running these tools on the full four-year BTC dataset reproduces the 4307.12% return result.

Table 7: Live Demo — Tool Execution Results

tablehead

1	`explain_strategy`	Completed	Yes
2	`analyze_performance`	Completed	Yes
3	`evaluate_risk`	Completed	Yes
4	`generate_signal`	Completed (FLAT)	Yes
5	`run_backtest`	Completed (0 trades)	Yes

10. Evaluation Against Judging Criteria

Table 8: Nasiko Judging Criteria Assessment

tablehead

Problem Value	**Strong**	Real quant use case; 4-year validated backtest; 4 stress tests; non-trivial analytical domain
Gateway Compliance	**Full**	Live-tested; all A2A fields correct; error envelopes implemented; AgentCard registered
Code Reliability	**Strong**	3-layer separation; full input validation in all tools; exception handling returns error strings never tracebacks
Output Predictability	**Full**	All 5 tools deterministic; zero stochastic elements; identical input always produces identical output

11. Forward Vision and Extensibility

AlphaQuant's architecture is deliberately extensible. Adding a new tool requires exactly four changes and leaves the executor and FastAPI layers untouched:

1. Add an `async def new_tool(...) -> str` method to `QuantTradingToolset`
2. Register it in `get_tools()`
3. Add a keyword cluster to `_detect_intent()`
4. Add a skill entry to `AgentCard.json`

11.1. Planned Extensions

- **Walk-forward validation** in `run_backtest` — rolling in-sample / out-of-sample windows to address the current purely in-sample limitation
- **Multi-asset comparison** via a `compare_assets` tool running parallel backtests and returning side-by-side metrics
- **ML-based intent classification** replacing keyword routing for more robust handling of ambiguous queries
- **Streaming signal updates** in `generate_signal` for real-time tick-by-tick signal generation

12. Conclusion

AlphaQuant demonstrates that an AI agent can be substantially more than a language model wrapper. By encapsulating the complete Impulse Flow Alpha strategy — its signal logic, backtest engine, stress test findings, and risk analysis — inside a modular, A2A-compliant service, AlphaQuant makes quantitative strategy analysis accessible, queryable, and actionable through plain English.

The system is **clean in its code** — three separated layers, each with a single responsibility. It is **precise in its outputs** — all metric computations are numerically identical to the original backtester notebook. It is **compliant in its protocol** — every response was verified live against the Nasiko gateway. And it is **honest in its limitations** — forward-looking risks are documented explicitly in the `evaluate_risk` tool and the forward vision section.

AlphaQuant was built to last beyond a hackathon submission. It is a genuine foundation for production quantitative strategy analysis.

Team Codex

Indian Institute of Technology (BHU), Varanasi

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